DATABASES
RELATIONAL MODEL

Relational Algebra

Konstantinos Katsifis

Relational Algebra

MEDITERRANEAN COLLEGE

After completing this chapter, you should be able to

- enumerate and explain the operations of relational algebra
  - (there is a core of 5 relational algebra operators),
- write relational algebra queries of the type join–select–project,
- discuss correctness and equivalence of given relational algebra queries.

MEDITERRANEAN COLLEGE

Overview

1. Introduction; Selection, Projection
2. Cartesian Product, Join
3. Set Operations
4. Outer Join

Example Database

MEDITERRANEAN COLLEGE

STUDENTS

SID	FIRST	LAST	EMAIL
101	Ann	Smith	...
102	Michael	Jones	(null)
103	Richard	Turner	...
104	Maria	Brown	...

EXERCISES

CAT	ENO	TOPIC	MAXPT
H	1	Rel.Alg.	10
H	2	SQL	10
M	1	SQL	14

RESULTS

SID	CAT	ENO	POINTS
101	H	1	10
101	H	2	8
101	M	1	12
102	H	1	9
102	H	2	9
102	M	1	10
103	H	1	5
103	M	1	7

Relational Algebra (1)

MEDITERRANEAN COLLEGE

- Relational algebra (RA) is a query language for the relational model with a solid theoretical foundation.
- Relational algebra is *not* visible at the user interface level (not in any commercial RDBMS, at least).
- However, almost any RDBMS uses RA to represent queries internally (for query optimization and execution).
- Knowledge of relational algebra will help in better understanding SQL and relational database systems in general.

Relational Algebra (2)

MEDITERRANEAN COLLEGE

- One particular operation of relational algebra is selection.

  Many operations of the relational algebra are denoted as greek letters. Selection is σ (sigma).

- For example, the operation $\sigma_{SID=101}$ selects all tuples in the input relation which have the value 101 in column SID.

Relational algebra: selection

$\sigma_{SID=101}$ (RESULTS)

RESULTS

SID	CAT	ENO	POINTS
101	H	1	10
101	H	2	8
101	M	1	12
102	H	1	9
102	H	2	9
102	M	1	10
103	H	1	5
103	M	1	7

=

SID	CAT	ENO	POINTS
101	H	1	10
101	H	2	8
101	M	1	12

Relational Algebra (3)

MEDITERRANEAN COLLEGE

- Since the output of any RA operation is some relation R again, R may be the input for another RA operation.

  The operations of RA nest to arbitrary depth such that complex queries can be evaluated. The final results will always be a relation.

- A query is a term (or expression) in this relational algebra.

A query

$$\pi_{\text{FIRST,LAST}}(\text{STUDENTS} \bowtie \sigma_{\text{CAT}='\text{M}'}(\text{RESULTS}))$$

Relational Algebra (4)

MEDITERRANEAN COLLEGE

- There are some difference between the two query languages RA and SQL:
  - Null values are usually excluded in the definition of relational algebra, except when operations like outer join are defined.
  - Relational algebra treats relations as sets, i.e., duplicate tuples will never occur in the input/output relations of an RA operator.

    Remember: In SQL, relations are multisets (bags) and may contain duplicates. Duplicate elimination is explicit in SQL (SELECT DISTINCT).

Selection (1)

MEDITERRANEAN COLLEGE

Selection

The selection σ_φ selects a subset of the tuples of a relation, namely those which satisfy predicate φ. Selections acts like a filter on a set.

Selection

$$\sigma_{A=1}\left(\begin{array}{|c|c|}\hline A & B \\ \hline 1 & 3 \\ 1 & 4 \\ 2 & 5 \\ \hline\end{array}\right) = \begin{array}{|c|c|}\hline A & B \\ \hline 1 & 3 \\ 1 & 4 \\ \hline\end{array}$$

Selection (2)

MEDITERRANEAN COLLEGE

- A simple selection predicate φ has the form

  (*Term*) (*ComparisonOperator*) (*Term*).

- (*Term*) is an expression that can be evaluated to a data value for a given tuple:
  - ✓ an attribute name,
  - ✓ a constant value,
  - ✓ an expression built from attributes, constants, and data type operations like $+, -, *, /$.

Selection (3)

MEDITERRANEAN COLLEGE

- (*ComparisonOperator*) is
  - = (equals), /= (not equals),
  - < (less than), > (greater than), ” , ≥,
  - or other data type-dependent predicates (*e.g.*, LIKE).
- Examples for simple selection predicates:
  - LAST = 'Smith'
  - POINTS ≥ 8
  - POINTS = MAXPT.

Selection (4)

MEDITERRANEAN COLLEGE

A few corner cases

$$\sigma_{C=1}\left(\begin{array}{|c|c|}\hline A & B \\ \hline 1 & 3 \\ 1 & 4 \\ 2 & 5 \\ \hline\end{array}\right) = \lightning \quad \text{(schema error)}$$

$$\sigma_{A=A}\left(\begin{array}{|c|c|}\hline A & B \\ \hline 1 & 3 \\ 1 & 4 \\ 2 & 5 \\ \hline\end{array}\right) = \begin{array}{|c|c|}\hline A & B \\ \hline 1 & 3 \\ 1 & 4 \\ 2 & 5 \\ \hline\end{array}$$

$$\sigma_{1=2}\left(\begin{array}{|c|c|}\hline A & B \\ \hline 1 & 3 \\ 1 & 4 \\ 2 & 5 \\ \hline\end{array}\right) = \begin{array}{|c|c|}\hline A & B \\ \hline\end{array}$$

Selection (5)

MEDITERRANEAN COLLEGE

- $\sigma_{\varphi}(R)$ corresponds to the following SQL query:

```
SELECT  *
FROM  R
WHERE φ
```

- A different relational algebra operation called projection corresponds to the SELECT clause. Source of confusion.

Selection (6)

MEDITERRANEAN COLLEGE

- More complex selection predicates may be performed using the Boolean connectives:

  $\varphi_1 \wedge \varphi_2$ ("and"), $\varphi_1 \vee \varphi_2$ ("or"), $\neg \varphi_1$ ("not").

- Note: $\sigma_{\varphi_1 \wedge \varphi_2}(R) = \sigma_{\varphi_1}(\sigma_{\varphi_2}(R))$.
- The selection predicate must permit evaluation for each input tuple in isolation. A predicate may not refer to other tuples.

Projection (1)

MEDITERRANEAN COLLEGE

Projection

The **projection** π_L eliminates all attributes (columns) of the input relation but those mentioned in the **projection list** L.

Projection

$$\pi_{A,C}\left(\begin{array}{|c|c|c|}\hline A & B & C \\ \hline 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \\ \hline\end{array}\right) = \begin{array}{|c|c|}\hline A & C \\ \hline 1 & 7 \\ 2 & 8 \\ 3 & 9 \\ \hline\end{array}$$

Projection (2)

MEDITERRANEAN COLLEGE

- The projection $\pi_{A_{i_1},\ldots,A_{i_k}}(R)$ produces for each input tuple $(A_1 : d_1, \ldots, A_n : d_n)$ an output tuple $(A_{i_1} : d_{i_1}, \ldots, A_{i_k} : d_{i_k})$.

  - π may be used to reorder columns.

    "σ discards rows, π discards columns."

  - DB slang: "All attributes not in L are projected away."

Projection (3)

- In general, the cardinalities of the input and output relations are not equal.

Projection eliminates duplicates

$$\pi_B \left(\begin{array}{|c|c|} \hline A & B \\ \hline 1 & 4 \\ 2 & 5 \\ 3 & 4 \\ \hline \end{array} \right) = \begin{array}{|c|} \hline B \\ \hline 4 \\ 5 \\ \hline \end{array}$$

Projection (4)

MEDITERRANEAN COLLEGE

- If RA is used to simulate SQL, the format of the projection list is often generalized:
  Attribute renaming:

$$\pi_{B_1 \leftarrow A_{i_1}, \ldots, B_k \leftarrow A_{i_k}}(R) \quad .$$

Computations (*e.g.*, string concatenation via || or arithmetics via +,-,...) to derive the value in new columns,

e.g.:

$$\pi_{\text{SID, NAME} \leftarrow \text{FIRST} \,||\, ' \,' \,||\, \text{LAST}}(\text{STUDENTS}) \quad .$$

Projection (5)

MEDITERRANEAN COLLEGE

- $\pi_{A_1,\ldots,A_k}(R)$ corresponds to the SQL query:

```
SELECT DISTINCT  A1, ..., Ak
FROM             R
```

- $\pi_{B_1 \leftarrow A_1,\ldots,B_k \leftarrow A_k}(R)$ is equivalent to the SQL query:

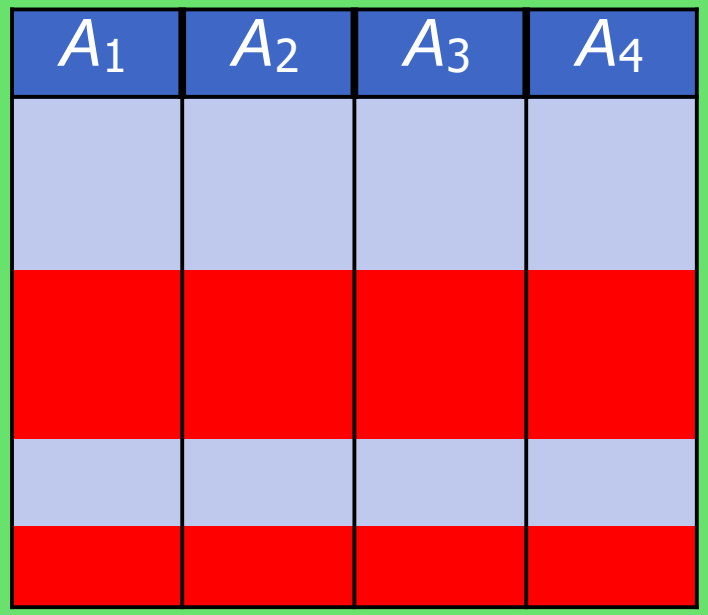
```
SELECT DISTINCT  A1 [AS] B1, ..., Ak [AS] Bk
FROM             R
```

Selection vs. Projection

MEDITERRANEAN COLLEGE

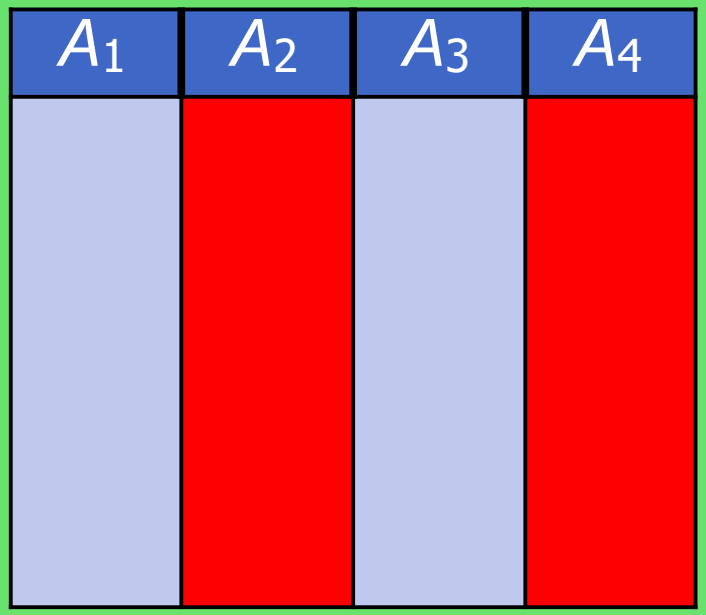
Selection vs. Projection

Selection σ



Filter some rows

Projection π



Projects all rows

Combining Operations (1)

MEDITERRANEAN COLLEGE

- Since the result of any relational algebra operation is a relation again, this intermediate result may be the input of a subsequent RA operation.
- Example: retrieve the exercises solved by student with ID 102:

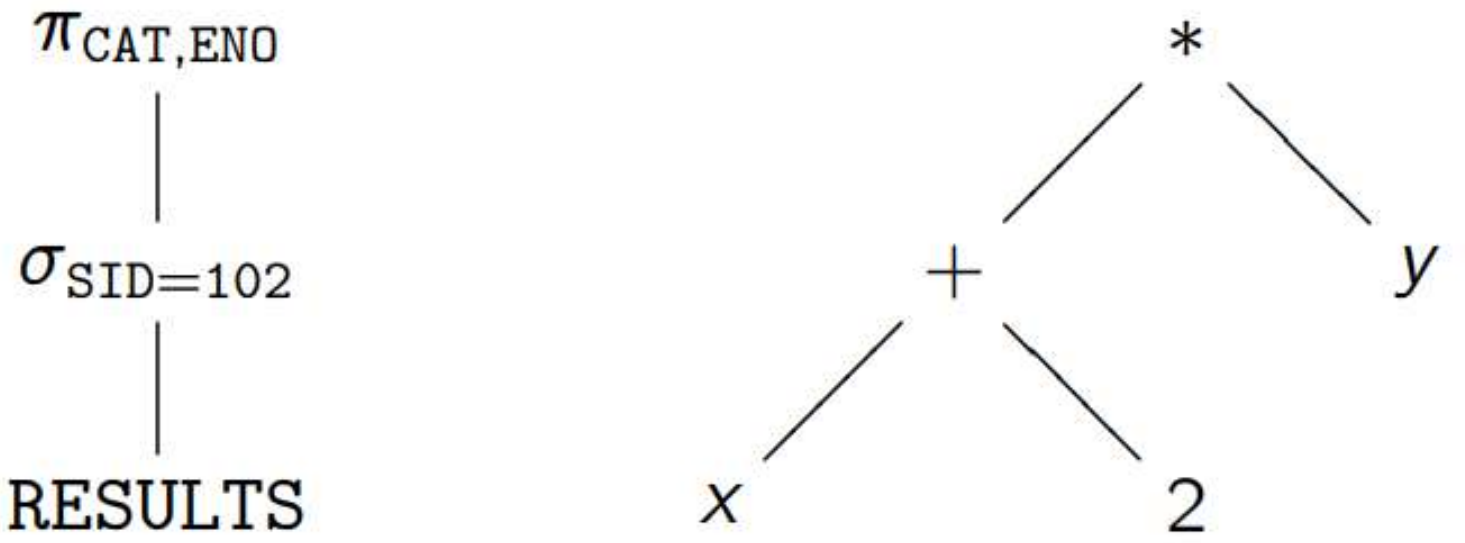
$$\pi_{\text{CAT,ENO}}(\sigma_{\text{SID=102}}(\text{RESULTS})) \ .$$

We can think of the intermediate result to be stored in a named temporary relation (or as a macro definition):

$$\text{S102} \leftarrow \sigma_{\text{SID=102}}(\text{RESULTS});$$
$$\pi_{\text{CAT,ENO}}(\text{S102})$$

Combining Operations (2)

- Composite RA expressions are typically depicted as operator trees:



- In these trees, computation proceeds bottom-up. The evaluation order of sibling branches is not pre-determined.

Combining Operations (3)

MEDITERRANEAN COLLEGE

- SQL permits the nesting of queries (the result of a SQL query may be used in a place of a relation name):

Nested SQL Query

```
SELECT DISTINCT CAT, ENO
FROM            (SELECT *
                 FROM   RESULTS
                 WHERE  SID = 102) AS S102
```

- Note that this is *not* the typical style of SQL querying.

Combining Operations (4)

MEDITERRANEAN COLLEGE

- Instead, a single SQL query is equivalent to an RA operator tree containing σ, π, and (multiple) $\times$ (see below):

SELECT-FROM-WHERE Block

```
SELECT DISTINCT CAT, ENO
FROM            RESULTS
WHERE           SID = 102
```

- Really complex queries may be constructed step-by-step (using SQL's **view** mechanism), S102 may be used like a relation:

SQL View Definition

```
CREATE VIEW S102
AS SELECT  *
   FROM    RESULTS
   WHERE   SID = 102
```

MEDITERRANEAN COLLEGE

Overview

1. Introduction; Selection, Projection
2. Cartesian Product, Join
3. Set Operations
4. Outer Join

Cartesian Product (1)

MEDITERRANEAN COLLEGE

- In general, queries need to combine information from several tables.
- In RA, such queries are formulated using $\times$, the Cartesian product.

Cartesian Product

The Cartesian product $R \times S$ of two relations R, S is computed by concatenating each tuple $t \in R$ with each tuple $u \in S$.

Cartesian Product (2)

MEDITERRANEAN COLLEGE

Cartesian Product

A	B
1	2
3	4

×

C	D
6	7
8	9

=

A	B	C	D
1	2	6	7
1	2	8	9
3	4	6	7
3	4	8	9

- Since attribute names must be unique within a tuple, the Cartesian product may only be applied if *R, S* do not share

any attribute names. (This is no real restriction because we have π.)

Cartesian Product and Renaming

MEDITERRANEAN COLLEGE

- $R \times S$ may be computed by the equivalent SQL query (SQL does not impose the unique column name restriction, a column A of relation R may uniquely be identified by $R.A$):

Cartesian Product in SQL

```
SELECT *
FROM   R, S
```

- In RA, this is often formalized by means of of a **renaming operator** $\varrho_X(R)$. If $sch(R) = (A_1 : D_1, \ldots, A_n : D_n)$, then

$$\varrho_X(R) \quad \equiv \quad \pi_{X.A_1 \leftarrow A_1, \ldots, X.A_n \leftarrow A_n}(R) \ .$$

Join (1)

MEDITERRANEAN COLLEGE

- The intermediate result generated by a Cartesian product may be quite large in general ($|R| = n, |S| = m \Rightarrow |R \times S| = n * m$).
- Since the combination of Cartesian product and selection in queries is common, a special operator join has been introduced.

Join

The **(theta-)join** $R \bowtie_\theta S$ between relations R, S is defined as

$$R \bowtie_\theta S \quad \equiv \quad \sigma_\theta(R \times S).$$

The **join predicate** θ may refer to attribute names of R and S.

Join (2)

MEDITERRANEAN COLLEGE

$\varrho_S(\text{STUDENTS}) \bowtie_{S.\text{SID}=R.\text{SID}} \varrho_R(\text{RESULTS})$

S.SID	S.FIRST	S.LAST	S.EMAIL	R.SID	R.CAT	R.ENO	R.POINTS
101	Ann	Smith	...	101	H	1	10
101	Ann	Smith	...	101	H	2	8
101	Ann	Smith	...	101	M	1	12
102	Michael	Jones	(null)	102	H	1	9
102	Michael	Jones	(null)	102	H	2	9
102	Michael	Jones	(null)	102	M	1	10
103	Richard	Turner	...	103	H	1	5
103	Richard	Turner	...	103	M	1	7

- Note: student Maria Brown does not appear in the join result.

Join (3)

MEDITERRANEAN COLLEGE

- Join combines tuples from two relations **and** acts like a filter: tuples without join partner are removed.

  Note: if the join is used to **follow a foreign key relationship**, then **no tuples are filtered:**

Join follows a foreign key relationship (dereference)

$$\texttt{RESULTS} \bowtie_{\texttt{SID}=S.\texttt{SID}} \pi_{S.\texttt{SID}\leftarrow\texttt{SID},\texttt{FIRST},\texttt{LAST},\texttt{EMAIL}}(\texttt{STUDENTS})$$

- There are join variants which act like **filters only:** left and right **semijoin** ($\ltimes$, $\rtimes$):

$$R \ltimes_\theta S \quad \equiv \quad \pi_{sch(R)}(R \bowtie_\theta S) \ ,$$

  or **do not filter** at all: **outer-join** (see below).

Natural Join

MEDITERRANEAN COLLEGE

- The **natural join** provides another useful abbreviation ("RA macro").

  In the natural join $R \bowtie S$, the join predicate θ is defined to be an **conjunctive equality comparison of attributes sharing the same name** in R, S.

  Natural join handles the necessary attribute renaming and projection.

Natural Join

Assume $R(A, B, C)$ and $S(B, C, D)$. Then:

$$R \bowtie S = \pi_{A,B,C,D}(\sigma_{B=B' \wedge C=C'}(R \times \pi_{B' \leftarrow B, C' \leftarrow C, D}(S)))$$

(**Note:** shared columns occur *once* in the result.)

Joins in SQL (1)

MEDITERRANEAN COLLEGE

- In SQL, $R \bowtie_\theta S$ is normally written as

Join in SQL ("classic" and SQL-92)

```
SELECT  *
FROM    R,S
WHERE   θ
```

or

```
SELECT  *
FROM    R JOIN S ON θ
```

- **Note:** the left query is **exactly** the SQL equivalent of $\sigma_\theta(R \times S)$ we have seen before.

SQL is a **declarative language:** it is the task of the SQL optimizer to infer that this query may be evaluated using a join instead of a Cartesian product.

Algebraic Laws (1)

MEDITERRANEAN COLLEGE

- A significant number **algebraic laws** hold for join which are heavily utilized by the query optimizer.
- Example: **selection push-down.**

  If predicate φ **refers to attributes in** S **only,** then

$$\sigma_{\varphi}(R \bowtie S) \quad \equiv \quad R \bowtie \sigma_{\varphi}(S) \ .$$

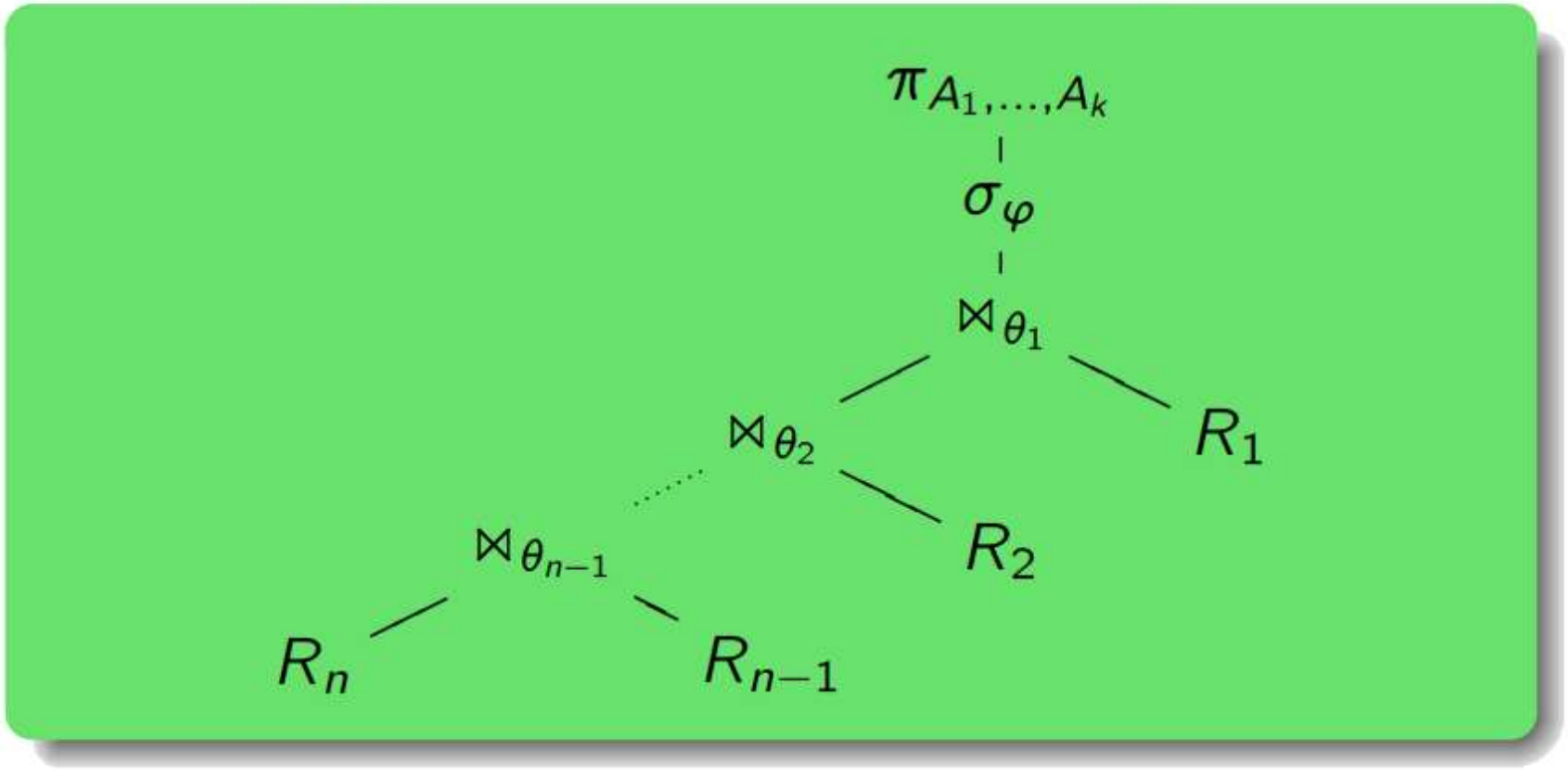
Selection push-down

Why is selection push-down considered one of the most significant algebraic optimizations?

A Common Query Pattern (1)

MEDITERRANEAN COLLEGE

- The following operator tree structure is **very** common:



① **Join** all tables needed to answer the query, ② **select** the relevant tuples, ③ **project** away all irrelevant columns.

A Common Query Pattern (2)

MEDITERRANEAN COLLEGE

- The select-project-join query

$$\pi_{A_1,\ldots,A_k}(\sigma_{\varphi}(R_1 \bowtie_{\theta_1} R_2 \bowtie_{\theta_2} \cdots \bowtie_{\theta_{n-1}} R_n))$$

has the obvious SQL equivalent

```
SELECT      A1, ..., Ak
DISTINCT FROM   R1, ..., Rn
WHER        φ
E AND       θ1 AND ... AND θn–1
```

- It is a common source of errors to forget a join condition: think of the scenario $R(A, B)$, $S(B, C)$, $T(C, D)$ when attributes A, D are relevant for the query output.

MEDITERRANEAN COLLEGE

Overview

1. Introduction; Selection, Projection
2. Cartesian Product, Join
3. Set Operations
4. Outer Join

Set Operations (1)

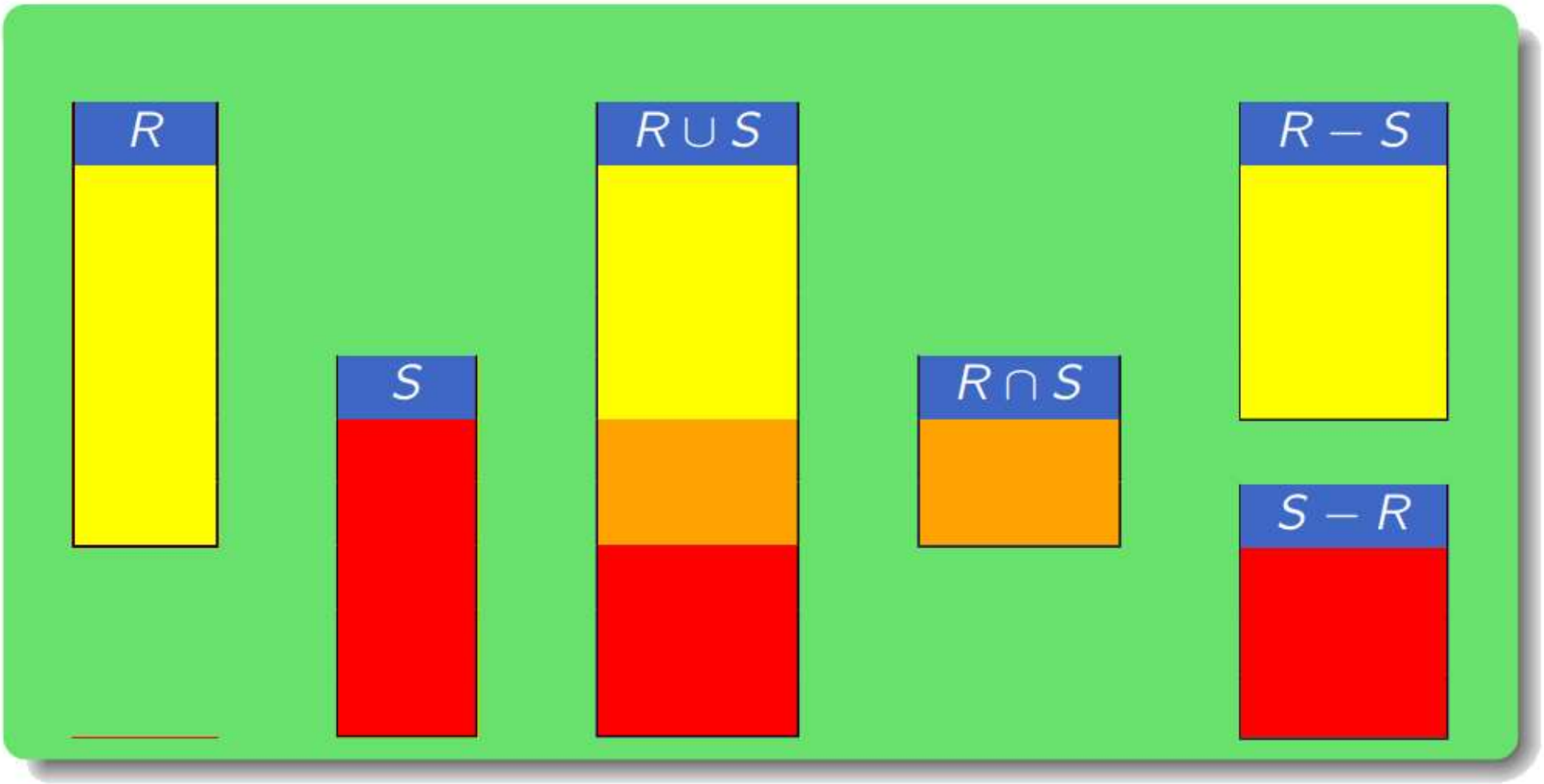
MEDITERRANEAN COLLEGE

- Relations are sets (of tuples). The "usual" family of binary set operations can also be applied to relations.
- It is a requirement, that both input relations have the same schema.

Set Operations

The set operations of relational algebra are $R \cup S$, $R \cap S$, and $R - S$ (union, intersection, difference).

Set Operations (2)



Union (1)

MEDITERRANEAN COLLEGE

- In RA queries, a typical application for $\cup$ is case analysis.

Example: Grading

$$\begin{aligned}
\texttt{MPOINTS} := {} & \pi_{\texttt{SID,POINTS}}(\sigma_{\texttt{CAT='M'} \wedge \texttt{ENO=1}}(\texttt{RESULTS})) \\
& \pi_{\texttt{SID,GRADE}\leftarrow\texttt{'A'}}(\sigma_{\texttt{POINTS} \geqslant 12}(\texttt{MPOINTS})) \\
\cup \quad & \pi_{\texttt{SID,GRADE}\leftarrow\texttt{'B'}}(\sigma_{\texttt{POINTS} \geqslant 10 \,\wedge\, \texttt{POINTS} < 12}(\texttt{MPOINTS})) \\
\cup \quad & \pi_{\texttt{SID,GRADE}\leftarrow\texttt{'C'}}(\sigma_{\texttt{POINTS} \geqslant 7 \,\wedge\, \texttt{POINTS} < 10}(\texttt{MPOINTS})) \\
\cup \quad & \pi_{\texttt{SID,GRADE}\leftarrow\texttt{'F'}}(\sigma_{\texttt{POINTS} \leqslant 7}(\texttt{MPOINTS}))
\end{aligned}$$

Union (2)

MEDITERRANEAN COLLEGE

- In SQL, $\cup$ is directly supported: keyword `UNION`.

`UNION` may be placed between two `SELECT-FROM-WHERE` blocks:

SQL's `UNION`

```
  SELECT SID, 'A' AS GRADE
  FROM   RESULTS
  WHERE  CAT = 'M' AND ENO = '1' AND POINTS >= 12
UNION
  SELECT SID, 'B' AS GRADE
  FROM   RESULTS
  WHERE  CAT = 'M' AND ENO = '1'
  AND    POINTS >= 10 AND POINTS < 12
UNION
  ...
```

Set Difference (1)

MEDITERRANEAN COLLEGE

- Note: the RA operators $\sigma, \pi, \times, \cap, \cup$ are monotic by definition, *e.g.*:

$$R \subseteq S \quad \Rightarrow \quad \sigma_\varphi(R) \subseteq \sigma_\varphi(S) \ .$$

- Then it follows that every query Q that exclusively uses the above operators behaves monotonically:
  - Let I_1 be a database state, and let $I_2 = I_1 \cup \{t\}$ (database state after insertion of tuple t).
  - Then every tuple u contained in the answer to Q in state I_1 is also contained in the anser to Q in state I_2.

  Database insertion never invalidates a correct answer.

Example Database (recap)

MEDITERRANEAN COLLEGE

STUDENTS

SID	FIRST	LAST	EMAIL
101	Ann	Smith	...
102	Michael	Jones	(null)
103	Richard	Turner	...
104	Maria	Brown	...

EXERCISES

CAT	ENO	TOPIC	MAXPT
H	1	Rel.Alg.	10
H	2	SQL	10
M	1	SQL	14

RESULTS

SID	CAT	ENO	POINTS
101	H	1	10
101	H	2	8
101	M	1	12
102	H	1	9
102	H	2	9
102	M	1	10
103	H	1	5
103	M	1	7

Set Difference (2)

MEDITERRANEAN COLLEGE

A correct solution?

$$\pi_{\text{FIRST,LAST}}(\text{STUDENTS} \bowtie_{\text{SID}\neq\text{SID2}} \pi_{\text{SID2}\leftarrow\text{SID}}(\text{RESULTS}))$$

A correct solution?

$$\pi_{\text{SID,FIRST,LAST}}(\text{STUDENTS} - \pi_{\text{SID}}(\text{RESULTS}))$$

Correct solution!

Set Operations and Complex Selections

MEDITERRANEAN COLLEGE

- Note that the availability of $\cup$, $-$ (and $\cap$) renders complex selection predicates superfluous:

Predicate Simplification Rules

$$\sigma_{\varphi_1 \wedge \varphi_2}(Q) \stackrel{\rightarrow}{=} \sigma_{\varphi_1}(Q) \cap \sigma_{\varphi_2}(Q)$$

$$\sigma_{\varphi_1 \vee \varphi_2}(Q) = \sigma_{\varphi_1}(Q) \cup \sigma_{\varphi_2}(Q)$$

$$\sigma_{\neg \varphi}(Q) = Q - \sigma_{\varphi}(Q)$$

Summary

MEDITERRANEAN COLLEGE

The five basic operations of relational algebra are:

1. σ_{φ} **Selection**
2. π_{L} **Projection**
3. $\times$ **Cartesian Product**
4. $\cup$ **Union**
5. $-$ **Difference**

- **Derived** (and thus redundant) **operations:** **Theta-Join** $\bowtie_{\theta}$, **Natural Join** $\bowtie$, **Semi-Join** $\ltimes$, **Renaming** ϱ and **Intersection** $\cap$.